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Key Points:

- Excavation-induced fractures were reconstructed using the pneumatic interference tests
- Equivalent porous media and discrete fracture network approaches were used to invert the pressure responses
- The results provide information about fracture connectivity, hydraulic properties, and the geometry of the dominant structures

Supporting Information:

Supporting Information may be found in the online version of this article.

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High-Resolution Characterization of Excavation-Induced Fracture Network Using Continuous and Discrete Inversion Schemes

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Abstract The Meuse/Haute-Marne Underground Research Laboratory provides the location for the experiment designed to investigate the induced fracture network around open or sealed galleries and drifts. The objective of this study is to investigate and reconstruct the hydraulic properties and the geometry of the induced fracture network to improve the insights and validate the conceptual model of the induced fracture network due to the stress redistribution during tunnel excavations. Within the presented study, the cross-hole responses of the pneumatic tests were analyzed in the first step with an equivalent porous media—3D travel time-based tomographic approach. In the next step selected 2D profiles of the 3D model domain were inverted using a discrete fracture network inversion approach. The database of the tomographic analysis is based on 18 gas injection tests and 151 pressure interferences, which were recorded between nine closely spaced boreholes. The travel time-based inversion approach allowed for the reconstruction of the 3D gas diffusivity distribution between nine boreholes with high-resolution. The applied discrete fracture network inversion approach is based on a transdimensional Markov Chain Monte Carlo (MCMC) methodology and it operates with reversible model updates (jumps) that change the problem dimensions, that is, the number, length and position of fractures within the model domain after each iteration step. The synthesis of the results between the reconstructed 3D diffusivity tomogram and the 2D fracture tomograms improved the insights into the spatial geometry of the induced fracture network around galleries.

1. Introduction

Clay formations are desirable host rocks for radioactive waste repository due to their unique characteristics as a natural barrier, such as low hydraulic conductivity, high sorption capacity, small molecular diffusion coefficient, significant radionuclide retention capacity, and high self-sealing potential (Bossart et al., 2018). However, the construction of subterranean facilities, such as tunnels and drifts might form an interconnected fracture network surrounding the openings. The characteristics of these features are determined by rock properties, in-situ stress conditions, excavation methods, the geometry of the openings, and time (Armand et al., 2014). The fundamental concern about the long-term safety of such radioactive waste repositories is how the presence of induced fractures will influence the host rock integrity, that is, under what conditions do the induced fractures serve as a channel for radionuclide migration? To answer this question, it is necessary first to identify the geometry of the induced fractures and then characterize the THM behavior of these fractures over time (Bossart et al., 2019).

Characterization of induced fracture network is a challenging task due to the complexity and heterogeneity of the interconnected fracture network which results in variable pore pressure gradient (including suction), sharp changes in permeability properties, and stress redistribution. There exists a variety of in-situ characterization methods in clay formations (Bock et al., 2010; Bossart et al., 2019; Martin et al., 2004). Hydraulic and pneumatic tomography tests are among the most common and practical tools for the hydraulic characterization of the induced fractures. These techniques are based on the measurement of pressure changes in the monitoring intervals as a result of fluid or gas pressure anomalies (injection or production) in the source borehole interval. The induced fractures are typically filled with air (i.e., partially saturated), especially after excavation, hence pneumatic cross-hole tests are preferred. There have been various studies on the single- and cross-hole pneumatic injection tests to characterize fractured rocks (Huang et al., 1999; V. Vesselinov et al., 2000; Illman, 2014). The outcome of all these studies is the estimation of the hydraulic properties, however, the spatial assignment of these values is challenging. Vesselinov et al., as an example, performed a three-dimensional numerical inverse model

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for the interpretation of cross-hole pneumatic tests in unsaturated fractured tuff at the Apache Leap Research Site (ALRS), Arizona (V. V. Vesselinov et al., 2001a, 2001b). During their work, they were able to reconstruct a high-resolution geostatistical estimate of the distribution of air permeability and air-field porosity in the tested rock volume on a length scale of 1 m. In addition, geophysical methods such as ground positioning radar (GPR), seismic, or electrical resistivity tomography (ERT) allow for the reconstruction of the geometry of the induced fracture network, but the estimation of fracture connectivity and hydraulic properties of the fractures can not be estimated directly (De La Vaissière et al., 2014).

In the past, hydraulic and pneumatic tomography tests have proven to provide high-resolution images of the hydraulic properties of the subsurface which goes beyond the expressiveness of conventional hydraulic or pneumatic test analysis (e.g., Brauchler et al., 2013; Illman et al., 2010; Yeh & Liu, 2000). In this study, we propose an application of this technology to field data recorded in the Meuse/Haute-Marne Underground Research Laboratory for the characterization of the excavation-induced fractures network. The recorded pressure responses are evaluated using two different approaches, which are equivalent porous medium (EPM) and discrete fracture network (DFN). In the EPM approach, fractures are represented by high conductivity zones. The results would be then a blurred visualization of hydraulic properties (e.g., hydraulic conductivity or diffusivity) (Illman, 2014). Fracture statistics and the resolution of hydraulic properties can be improved by increasing the number of sources and receivers (Illman, 2014; Sharmeen et al., 2012). On the other hand, fractures can be treated explicitly as a discrete fracture network (DFN) and the hydraulic and structural properties are estimated directly. This method requires detailed specification of fracture geometry and their spatial distribution. In this study, we follow the approach proposed by Dong et al. (2019) and combine the EPM approach which in our case is the travel time-based tomographic approach (Brauchler et al., 2003, 2011; Vasco et al., 2000) and a DFN approach based on transdimensional reversible jump Markov Chain Monte Carlo algorithm (Jiménez et al., 2016; Somogyvári et al., 2017). Thereby, the results of the travel time-based inversion approach are used to constrain and interpret the DFN inversion results. This comparison is possible as both approaches aim for recognition of conductive fluid paths. These features are shown by higher diffusivity zones in EPM and higher fracture probability in the DFN approach.

2. Description of the Excavation-Induced Fractures at the Meuse/Haute-Marne URL Drifts

The Meuse/Haute-Marne Underground Research Laboratory (URL) is located in the Callovo-Oxfordian clay rock formation on the eastern edge of the Paris Basin in France. The URL has been constructed as part of ANDRA's research program to investigate the feasibility of radioactive waste disposal in a deep geologic deposit. Within the URL, the potential host rock for nuclear waste, the Callovo-Oxfordian clay rock is found between 420 and 550 m below ground overlain and underlain by poorly permeable carbonate formations (Armand et al., 2017; Delay et al., 2007). The current layout of the URL is shown in Figure 1. The experimental drift network is located at the main level (−490 m) and at −445 m of the URL (Figure 1). At the main level, the galleries are mainly oriented parallel or orthogonal to the horizontal principal stress direction. Armand et al. (2014) proposed a detailed conceptual model of the induced fracture network around galleries for the main level of the URL. Based on core, overcored resin-filled samples, geological survey of the drift face and sidewalls as well as the hydraulic conductivity estimates derived from multi-packer tests, a detailed and sound conceptual model of the induced fracture network considering the stress field orientation could be developed. It is worthwhile to mention here that all the fractures observed on cores are excavation-induced fractures and no tectonics fractures are encountered in all the cored boreholes at the URL.

The tomographic measurement data set was collected from nine closely spaced wells drilled in the tunnel floor in the GER gallery (Figure 1), which is oriented parallel to the minimum horizontal principal stress direction. In the following, we propose the associated conceptual model displayed in Figure 2. The conceptual model indicates that the extension of the induced fracture network is larger in the ceiling and floor of the gallery than the sidewalls. Focusing on the tunnel floor it can be observed that the lower chevron fractures can be described as moderately inclined structures that dip in the opposite direction of the excavation advance direction. Armand et al. (2014) describe two different zones: zone (a) is mainly dominated by (interconnected) extension fractures ranging from the tunnel floor to a depth of around 1.2 m, which is equivalent to 30% of the tunnel diameter; zone (b) is characterized by (discrete) shear fractures ranging to a depth of 3.7 m, which is equivalent of 0.8 of

Location of nine borehole at the GER gallery floor

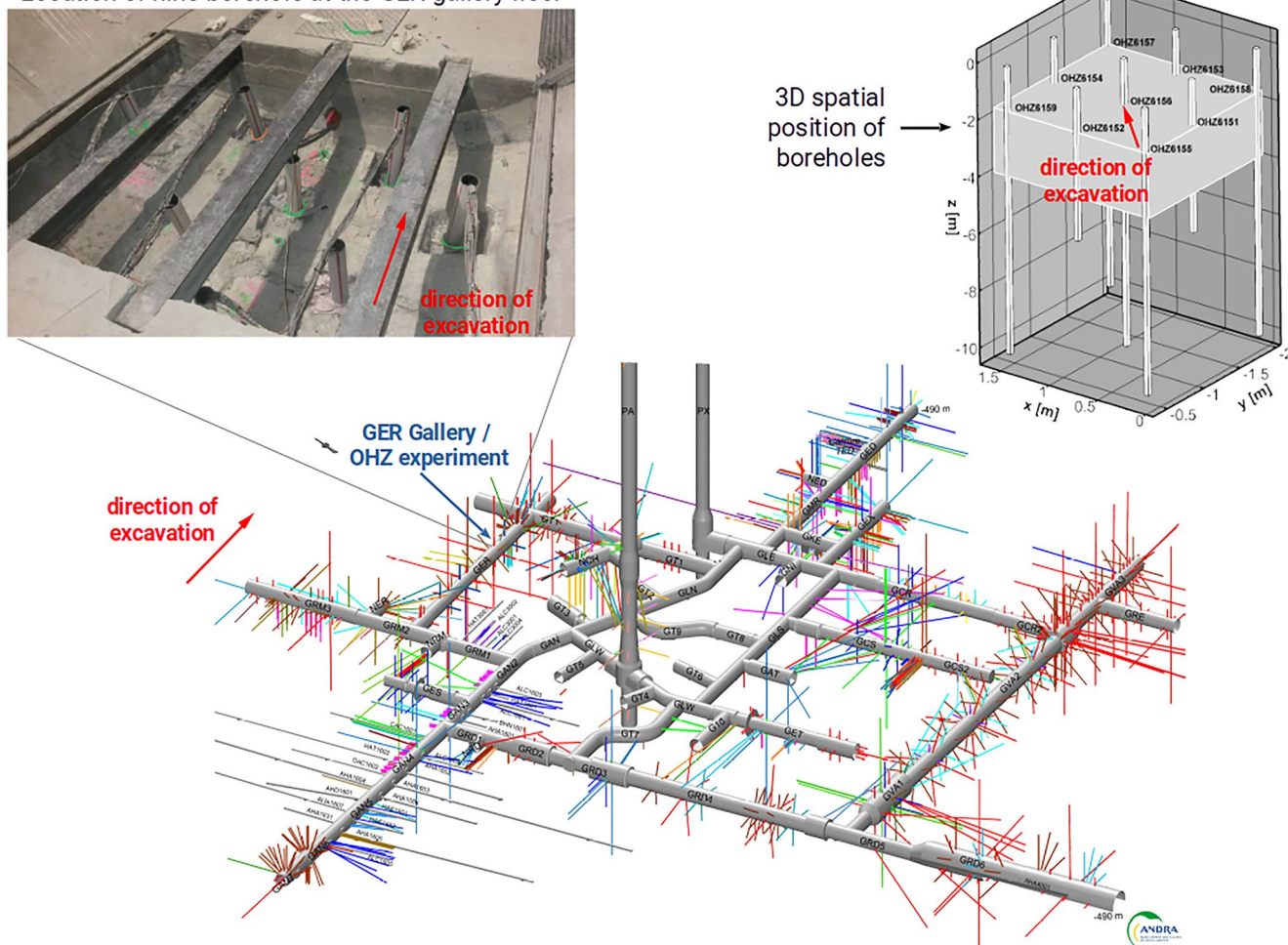


Figure 1. Overview of the galleries in the Meuse/Haute-Marne Underground Research Laboratory. The location of nine boreholes in the GER gallery floor is shown with respect to the direction of tunnel excavation.

the tunnel diameter. The goal of this work is to study and reconstruct the hydraulic properties and the geometry of the induced fracture network, particularly the lower chevron structure, in order to improve the insights and validate the conceptual model of the fracture network as a function of the stress field.

3. Experimental Data Set

Cross-hole pneumatic injection tests were performed on the tunnel floor of the GER gallery among nine boreholes. The GER gallery is oriented parallel to the minimum horizontal principal stress direction. Each borehole is equipped with a multi-packer system consisting of two intervals that can be used for injection and/or observation. For further information about the applied equipment can be found in Fierz et al. (2007). The intervals located closer to the tunnel floor are labeled as position (a) and the intervals further away are labeled as position (b). The length of the intervals and their relative position to the drift wall is given in Table 1. The distances between the nine boreholes vary between approximately 0.7 and 2.0 m. The nine boreholes and the relative position of the boreholes to each other are shown in Figure 1.

The gas injection rates for the constant rate tests performed in position (a) of the boreholes OHZ6151 to OHZ6159 varied between 18 and 29 l/min and the injection tests were performed for a period between 3 and 70 min. For the tests performed in the intervals (b) in position (b) for the boreholes OHZ6151 to OHZ6159, the injection rates were one order of magnitude smaller and varied between 0.15 and 2.84 l/min and the gas injection tests were carried out for a period between 10 and 60 min (Table 1). The larger injection rate and the smaller injection

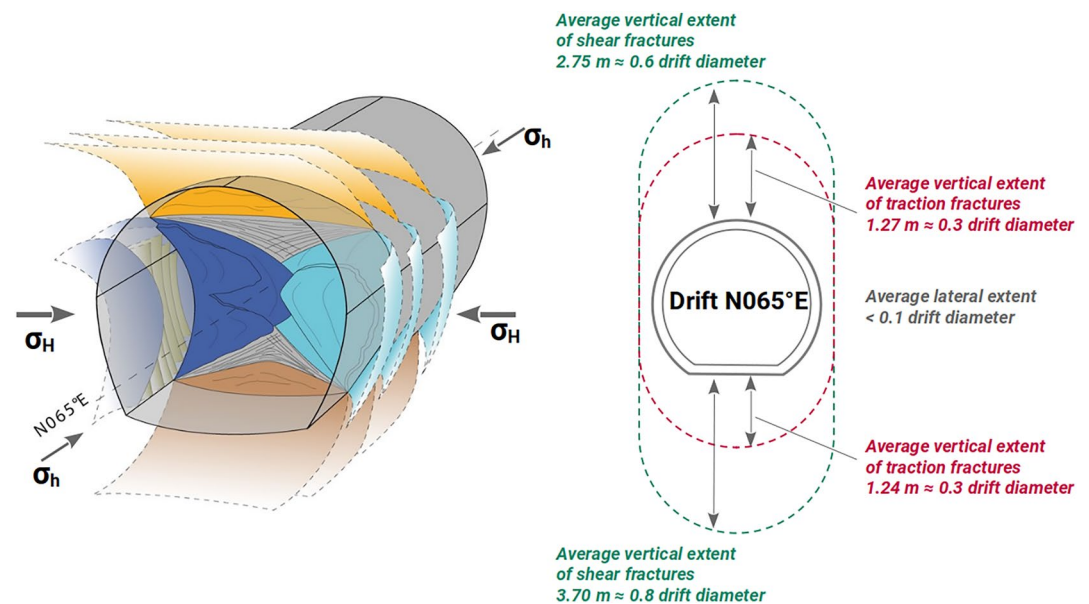


Figure 2. Conceptual model of the induced fractures network around a gallery parallel to the minimum horizontal principal stress direction after Armand et al. (2014).

time of the constant rate tests performed in the intervals at position (a) in comparison with the tests performed in the intervals at position (b) can be explained by the fact that the zone closer to the tunnel floor is characterized by larger fractures mainly extension fractures than farther away where the fracture frequency of the extension fracture is decreasing and mainly shear fractures occur. This pattern can be validated by the results derived from a numerical analysis of the single borehole pressure data recorded in the respective test intervals and displayed in Table 1.

Table 1
Description of the Nine Boreholes in GER Gallery

Borehole	Interval	Depth [m]	ΔP [bar]	Injection time	Injection rate [lit/min]	Gas conductivity [m/s]
OHZ 6151	a	1.31–2.41	0.07	15	28.48	2.8×10^{-6}
	b	3.11–4.21	0.53	60	1.65	1.7×10^{-8}
OHZ 6152	a	1.31–2.41	0.02	3	29.05	9.7×10^{-6}
	b	3.11–4.21	0.38	60	1.95	1.9×10^{-8}
OHZ 6153	a	1.31–2.41	0.12	12	28.70	2.1×10^{-6}
	b	3.11–4.21	0.60	30	0.25	2.4×10^{-9}
OHZ 6154	a	1.31–2.41	0.01	10	26.70	1.5×10^{-5}
	b	3.11–4.21	0.40	35	0.55	2.7×10^{-9}
OHZ 6155	a	1.34–2.44	0.46	70	17.93	5.1×10^{-7}
	b	3.14–4.24	0.70	60	0.15	6.8×10^{-10}
OHZ 6156	a	1.36–2.46	0.26	25	26.60	1.1×10^{-6}
	b	3.16–4.26	0.45	10	0.80	6.2×10^{-9}
OHZ 6157	a	1.35–2.45	0.70	20	25.24	5.1×10^{-7}
	b	3.15–4.25	0.50	60	1.10	1.2×10^{-8}
OHZ 6158	a	1.20–2.30	0.41	14	26.70	4.4×10^{-7}
	b	3.00–4.10	0.41	35	1.10	1.3×10^{-8}
OHZ 6159	a	1.20–2.30	0.07	20	26.58	3.0×10^{-6}
	b	3.00–4.10	0.36	60	2.84	4.7×10^{-8}

The analysis was performed using the borehole simulator MULTISIM, an in-house development of the company AFRY. The borehole simulator is adapted to the requirements of hydraulic and pneumatic test analysis in low permeable media (Tauzin & Johns, 1997). More information about the MULTISIM simulator can be found in the Supporting Information. The simulator is based on a 1D radial symmetrical homogeneous model and allows the numerical estimation of the hydraulic and pneumatic parameters in low permeable media. For the borehole simulator analysis as well as for the analysis performed with the inversion approaches described in the following sections, we assume single-phase flow through the fracture network. This assumption might be violated in some distance to the gallery wall, that is, in zone b in which parts of the matrix might be partly saturated. However, this simplification can be valid as the main goal of the investigation is to characterize the fracture network with focus on the extension fractures occurring closer to the gallery wall. Bear in mind that pressure propagation is dominated by the fracture network. Brauchler et al. (2003) have realized a series of gas flow cross-well tests with pressure differences between 0.3 and 0.5 bar. They observed no influence on the shape of the observed pressure curves. In our example the applied pressure difference for test intonation is below 0.5 bar, hence, we conclude that changes in the gas compressibility can be neglected. The borehole storage effects can have a considerable effect on the observed pressure curve at the observation interval (Brauchler et al., 2007). In order to minimize these effects, the interval volume was strongly reduced by inserting a volume reducer element consisting of duroplastic in the injection and observation intervals, which led to an interval volume of 0.001 m³.

The results show that the estimated gas permeability values for the test intervals in zone (a)—an area closer to the tunnel surface—are two orders of magnitude larger than the gas permeability estimated for zone (b). Note a porosity value of 15% was assumed for both zones. This value was derived from the analysis of the pressure curves in 18 pneumatic tests using the borehole simulator MULTISIM. Thereby, the porosity value of 15% reflects the average value of the matrix and the fracture network.

In total, 18 pneumatic cross-hole tests were performed and 306 interferences are theoretically available for the inversion, whereby 153 independent interferences were selected for the inversion based on the following criteria:

- Reciprocal interferences were compared to each other and the ones with better signal-to-noise ratio were chosen (153 out of 306 interferences). The reasoning behind this filtering is that the reciprocity is usually not applicable to the field data in contrast to the theoretical and synthetic studies (Bruggeman, 1972) due to various reasons such as varying borehole conditions and imperfections.
- For pressure interferences that are propagating from packer intervals located at position (a) to packer intervals located at position (b) or vice-versa, we have always chosen the interference propagating from the zone (a) to the zone (b), because a much higher flow rate was applied than for the gas tests, which were initiated in the zone (b) (Table 1). Hence, the signal-to-noise ratio is much higher for interferences traveling from packer intervals located at position (a) to packer intervals located at position (b) than for the interferences traveling in the opposite direction.

Interferences, which were measured along the borehole are affected by the properties of the borehole damaged zone (bdz). For this reason, for each travel time, which is derived from an interference measured along the borehole, the following procedure was applied:

- If the travel time measured along the borehole is much smaller in comparison with the travel times derived from interferences (observed in another borehole) with the same source, the travel time for the inversion is not used. The chance is high that such a travel time only reflects the properties of the bdz.
- If the travel time measured along the borehole is larger or comparable with the travel times derived from interferences with the same source, the travel time was used for the inversion. In this case, the bdz is not characterized by a higher permeability than the induced fracture network.

Finally, 151 interferences were selected for the travel time-based inversion approach to reconstruct a 3D diffusivity tomogram and for the 2D DFN inversion between seven and twelve interferences were used depending on the reconstructed profile.

4. Travel Time-Based Inversion

Within the presented study, the cross-hole responses of the pneumatic injection tests were analyzed in the first step with an equivalent porous media—3D travel time-based tomographic approach. In a second step, selected

2D profiles of the 3D model domain were inverted using a discrete fracture network inversion approach, whereby the insights gained from the travel time-based inversion approach were used to constrain and interpret the discrete fracture network inversion results as suggested by Dong et al. (2019).

4.1. Travel Time-Based Inversion Approach

The proposed travel time inversion is based on the work of Vasco et al. (2000), who developed an inversion scheme based on the transformation of the groundwater flow equation into the eikonal equation (Virieux et al., 1994), which can then be solved in a computationally efficient way using ray tracing or particle tracking methods. The key element of the procedure is the following line integral relating the arrival time of a *hydraulic signal* to the reciprocal value of the hydraulic diffusivity (Vasco et al., 2000):

$$\sqrt{t_{peak}} = \frac{1}{\sqrt{6}} \int_{x_1}^{x_2} \frac{ds}{\sqrt{D(s)}} \quad (1)$$

where t_{peak} is defined as the source-to-observation-point travel time of the peak of a transient pressure curve, resulting from a Dirac signal generated at point x_1 (source), traveling along the propagation path s , and recorded at point x_2 (observation point). D is the hydraulic diffusivity. In our study, diffusivity is reconstructed by utilizing the GEOTOM3D, which is based on the SIRT (Simultaneous Iterative Reconstruction Technique) and ray tracing algorithm originally developed to determine seismic velocities. The curved ray tracing algorithm is described in great detail by Brauchler et al. (2013). The complete inversion procedure delivers a voxel-based 3D image of the diffusivity distribution between the nine boreholes.

Equation 1 is converted to a discrete problem by approximating the line integrals with a summation to solve the inverse problem. Thus, the goal of discretization is to adapt the number of cells or voxels to the number of measurements. In a lot of experiments, this practice cannot be achieved as the trajectory density depends on the spatial distribution of the injection and observation intervals and the parameter distribution itself. Consequently, the trajectory density is different in each cell. However, the regularly arranged position of the boreholes and the associated intervals, displayed by red discs in Figure 4a, minimize the influence of a varying trajectory density on the inversion.

The model domain consists of six cells in the x - and y -direction and four cells in the z -direction, which results in a mesh consisting of 144 cells. The cell length in x - and y -direction is 0.26 m and in the z -direction is 0.57 m. This results in a quadratic model domain with an edge length of 1.56 m in the x - and y -direction with a thickness of 2.28 m.

In order to increase the nominal resolution of the coarse grid, the staggered grid method developed by Vesnaver and Böhm (2000) was utilized. The method combines the advantages of low-resolution grids (stable inversion) with the need for high-resolution information. The method is based on the displacement of the initial grid in all directions. The displacement rates Δx , Δy , Δz are 1/36 of the voxel length in x - and y - and 1/24 of the voxel height in z -directions, respectively. For each grid position, a slightly different image was obtained, because inside each voxel the interstitial velocity over a different volume was averaged. The final mesh is composed of 31,104 cells, 36 in the x -direction, 36 in the y -direction and 24 in the z -direction and is displayed in Figure 4a.

4.2. Data Processing for the Travel Time-Based Inversion

As already mentioned in the Section 4.1, Equation 1 is only valid for an impulsive type source (Dirac source). By differentiation of transient head data measured at the receiver(s), the inversion scheme is also applicable to experiments with constant pumping rates over a given short period of time (Brauchler et al., 2010; Hu et al., 2011).

The estimation of the time derivative of noise-free data, for example, synthetic data, is straightforward. However, estimating the time derivative of field data can present a challenge due to measurement errors and limits in the resolution of the pressure transducers. In this study, the following three steps to define the first time-derivation of the measured pressure curves recorded in the observation intervals were performed:

1. Wavelet de-noising Figure 3a,
2. A polynomial regression is applied to the smoothed transient pressure data Figure 3b,
3. The first derivative of the polynomial fit is derived analytically Figure 3c.

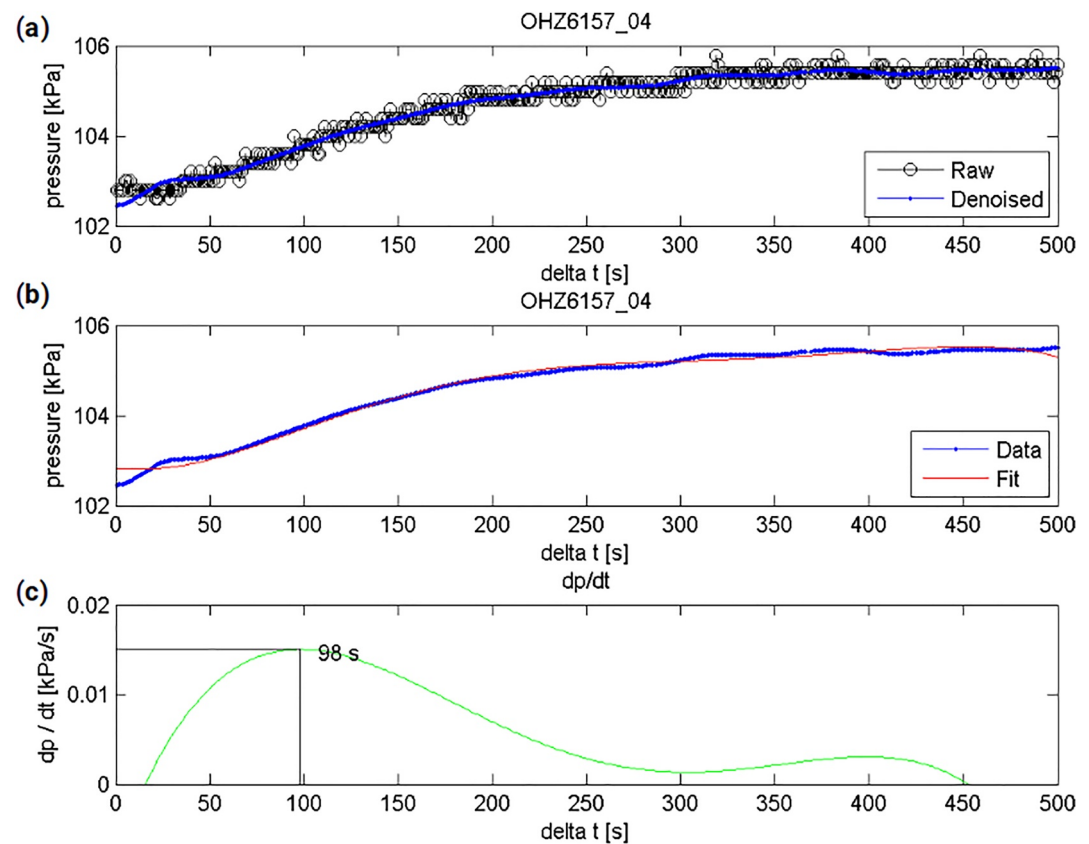


Figure 3. Time-derivation steps of the measured pressure data: (a) wavelet de-noising, (b) polynomial fitting, (c) first derivative of the polynomial fit.

Further information on the wavelet de-noising can be found in Yang et al. (2020).

4.3. Results—Diffusivity Tomograms

Figures 4b and 4d shows the diffusivity tomograms recorded among the boreholes OHZ6151–OHZ6159. The range of the reconstructed diffusivity varies between $1 \times 10^{-1} \text{ m}^2/\text{s}$ and $4 \times 10^{-4} \text{ m}^2/\text{s}$. The highest diffusivity values were close to the tunnel floor, orthogonal to the excavation along a half circle between boreholes OHZ6158 and OHZ6159 (Figure 4d). With increasing distance from the tunnel floor, the diffusivity values are decreasing by more than two orders of magnitude. For better visualization, we displayed slices parallel to the tunnel axis (Figure 4c) and slices orthogonal to the tunnel axis (Figure 4e).

The slice parallel to the tunnel axis can be described as follows: the highest diffusivity values in each depth can be roughly connected by a concave curve (dotted line in Figure 6c). This is in agreement with the shape of the chevron structures (Figure 2). The slices orthogonal to the tunnel axis show the following: the slice with the greatest distance from borehole OHZ6155 is characterized by the highest diffusivity value in the upper part of the model domain of all slices (Figure 4e, left). The slice with the smallest distance from borehole OHZ6155 (Figure 4e, right) is characterized by a higher diffusivity value in the lower part of the mode domain. This indicates that the preferential flow path is dipping in the opposite direction of the excavation advancement, which is in agreement with the shape of the chevron structures. Note, generally the diffusivity is decreasing with increasing distance from the tunnel floor.

Due to the large number of interferences and the regular distribution of test and observation intervals, the whole model domain is characterized by a high trajectory density, which, means no cell exists that is not crossed by four or more trajectories. Hence, no areas with limited significance with respect to the hydraulic parameter reconstruction have to be covered or hidden, respectively.

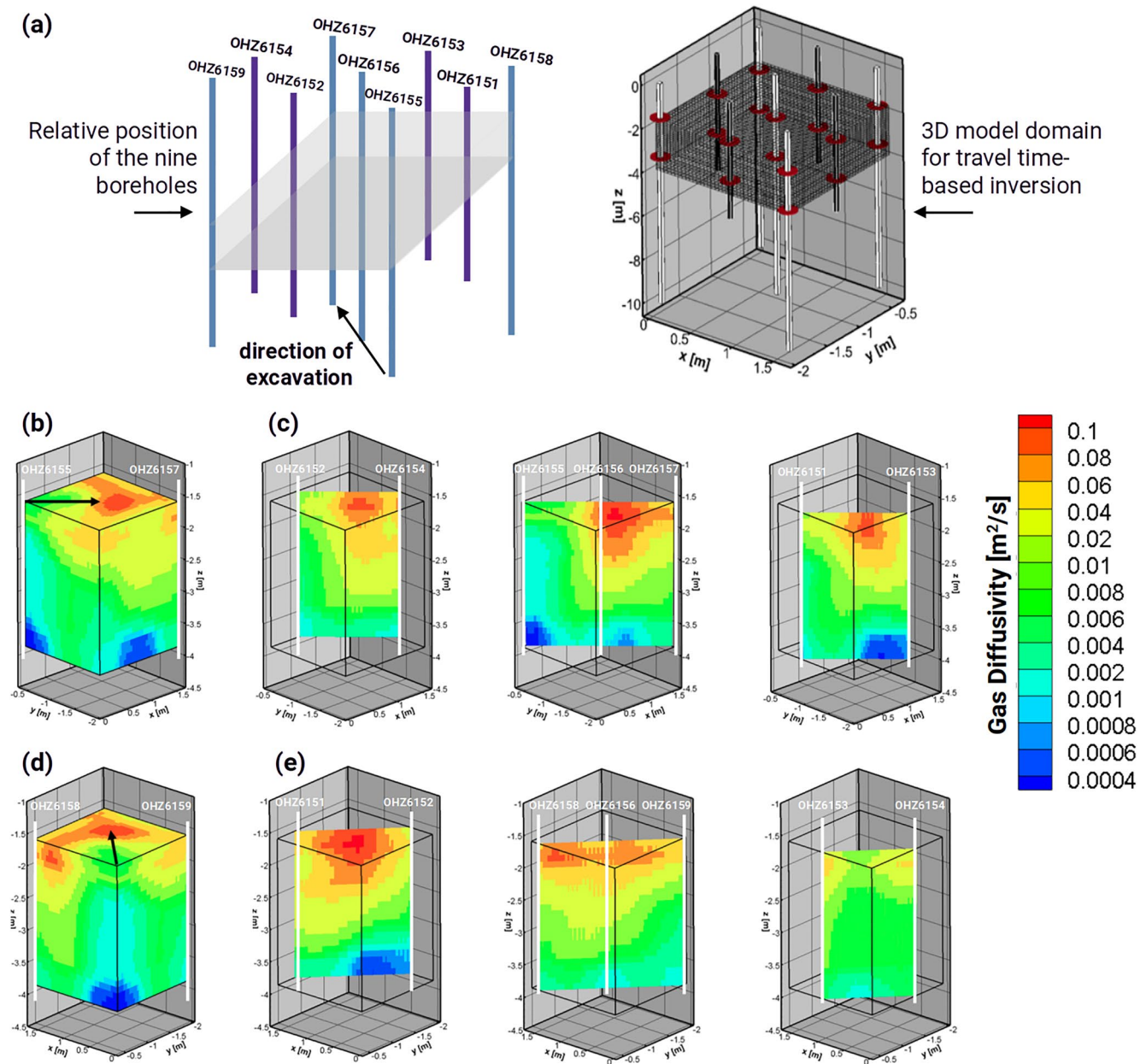


Figure 4. 3D diffusivity tomogram: (a) spatial position of boreholes and intervals (the location of intervals are shown by red discs), (b) 3D model domain with the black arrow indicating the tunnel axis direction, (c) vertical slices parallel to the tunnel axis, (d) rotated 3D model domain with the arrow indicating the tunnel axis direction, (e) vertical slices orthogonal to the tunnel axis.

The 3-D diffusivity tomograms displayed in Figure 4 show the potential of the applied travel time-based inversion technique to reconstruct changes in hydraulic properties in three dimensions with a high spatial resolution. This information is used in the next step to constrain the DFN inversion and to support the interpretation of the DFN results.

5. DFN-Based Inversion

5.1. DFN Inversion Approach

A Bayesian inversion framework is used to fit the DFN models to the observed pressure measurement (interferences) in the observation intervals. In general, Bayesian inference aims to provide a probabilistic model to a set of

unknown parameters and deliver a probabilistic characterization of unknown model parameters. Bayesian inversion has been applied to related geological problems such as the characterization of flow and transport in heterogeneous media (Lee & Kitanidis, 2014; Mondal et al., 2010) and reconstruction of the stress field (Lecampion & Lei, 2010). The Markov Chain Monte Carlo (MCMC) method is one of the most common techniques of Bayesian inference in geosciences. MCMC has been applied to solve inverse problems using various sources of information such as hydraulic tomography (Ringel et al., 2019, 2021), tracer tomography (Jiménez et al., 2016; Somogyvári et al., 2017), fracture network intersections with boreholes (Mardia et al., 2007), and in-situ stress state field data (Afshari Moein et al., 2018). The transdimensional reversible jump Markov Chain Monte Carlo (rjMCMC) is an extended variant of MCMC, in which the number of parameters can vary among subsequent iterations during the inversion process. Somogyvári et al. (2017) introduced this inversion approach for calibrating orientation, lengths, and numbers of fractures in DFNs using cross-well tracer tomography experiments. In this work, we adopt the approach of Somogyvári et al. (2017) to conduct gas pressure tomography inversion for the construction of the induced fracture network around the GER gallery in the Meuse/Haute-Marne URL.

In the proposed study, the examined model parameters are the geometrical properties of the induced fracture network around the excavated tunnel, that is, their locations and lengths. The other physical properties such as apertures are considered constant and uniform for each fracture set. The fracture length and orientation distributions are the key statistical properties of this work. A truncated fracture length distribution is considered for this study as the small fracture segments are less relevant for flow and transport (Baghbanan & Jing, 2007; Bour & Davy, 1997; Darcel et al., 2003; de Dreuzy et al., 2002). The inversion procedure is initiated by randomly generating a DFN realization (θ) based on given statistical properties. The rjMCMC is an iterative algorithm where each iteration consists of an update and an evaluation phase. In the update phase, a new DFN realization is proposed (θ') by sequentially updating the geometry of the previous one. For each update, three geometry modifications are possible, which are fracture addition, fracture deletion, and fracture shift (Somogyvári et al., 2017). The type of update is randomly selected according to initially defined probabilities. After the update, the evaluation phase starts with the simulation of the tomographic experiment using the proposed new DFN realization. The updated model was then evaluated using the Metropolis-Hastings-Green acceptance criteria (Brooks et al., 2011), which are constructed from the product of four terms: the prior, the likelihood and the proposal ratio and the Jacobian in Equation 2:

$$\alpha = \min \left[1, \frac{P(\theta')}{P(\theta)} \frac{L(\xi|\theta')}{L(\xi|\theta)} \frac{q(\theta' \rightarrow \theta)}{q(\theta \rightarrow \theta')} |J| \right] \quad (2)$$

The prior ratio $\left(\frac{P(\theta')}{P(\theta)} \right)$ is calculated from the prior distributions, but as we are using non-informative priors, this term equals to one. The likelihood ratio $\left(\frac{L(\xi|\theta')}{L(\xi|\theta)} \right)$ shows the improvement of the model over the update, calculated from the misfits between observed and simulated experiment data. Assuming a normal distribution for the observations, this term can be calculated from the root mean square (RMS) errors between the simulated and the observed measurements for the same source-receiver combinations. The update ratio $\left(\frac{q(\theta' \rightarrow \theta)}{q(\theta \rightarrow \theta')} \right)$ consists of the probability of the used update and its reverse counterpart. This term makes sure that updates with low probability are less likely to get accepted. The final term in the expression is the so-called Jacobian ($|J|$), a matrix determinant that controls any scale changes due to the dimensionality jumps. For transitions between discretized spaces (which is our case), this term is always one (Denison et al., 2002). Hence in our case, the acceptance criteria simplifies to the product of the likelihood and the proposal ratio. The acceptance ratio provides the probability with which the updated DFN realization can be accepted. If the proposed DFN is not accepted, the chain stays at the previous realization, while in case of acceptance, it moves to the updated DFN. After the evaluation phase, the next iteration starts.

The MCMC chain samples the posterior probability distribution of the inverse problem, and provides a representative set of DFN realizations called the ensemble as a result. The ensemble is a collection of equally probable DFN solutions. To remove the effect of the initial model choice (a) the chain needs to be run long enough to converge and (b) the first part of the chain needs to be discarded from the result. In our experience, chains of over 10,000 samples are sufficient for the presented applications.

5.2. Implementation of the DFN Inversion

The MCMC algorithm uses three possible updates to propose a new DFN realization: fracture addition, deletion and shift. Each of these updates is implemented in a way, that the MCMC chain remains reversible and the

probability of the updates stays controlled. The exact probabilities of the updates are detailed in the following subsections.

5.2.1. Fracture Addition

A fracture is added to the DFN, from one of the fracture sets selected randomly. The DFN is discretized with a pre-defined spacing distance, then one of these points is selected for the fracture insertion. Then the fracture length is drawn from a discrete fracture length distribution. This distribution may be truncated if the space around the insertion point is limited. We avoid fracture additions resulting in overlapping fractures, as our updates do not account for any fracture propagation. Lastly, the relative alignment between the added fracture and the insertion point is assigned. The final probability of the update is calculated from the probabilities of the substeps.

$$P_{add} = P_{set} P_{insertion} P_{length} P_{align} \quad (3)$$

5.2.2. Fracture Deletion

A randomly selected fracture is removed from the DFN. Floating fractures, fractures that are not connected to any other fracture cannot be deleted, because there would be no reversible fracture addition due to the previously defined addition rules. Additionally, connections between all source and receiver points need to be preserved to run the forward model properly. Hence, before the update, the deletable fractures need to be identified, and their number considered when calculating the probability of the update.

$$P_{del} = \frac{1}{count(deletable\ fractures)} \quad (4)$$

The reverse move of fracture addition is fracture deletion and vice versa. When calculating the proposal ratio, the probability of the reverse move needs to be calculated on the updated DFN.

5.2.3. Fracture Shift

Fracture shift is an n -dimensional update, where a selected fracture is shifted to another location. It is implemented as a deletion-addition step while maintaining the length of the fracture the same. The update should also preserve the connections between the source and receiver points. This is a symmetric update, equally probable in forward and reverse direction, so the proposal ratio is 1 in this case.

5.3. Data Preparation

In total, 18 pneumatic gas tests were conducted, and more than two hundred interferences were recorded. In total three 2D inversions were performed. The three profiles are aligned parallel to the excavation direction and cross the chevron fractures perpendicularly. For better visualization, we have overlain the spatial position of the 2D profiles with the reconstructed diffusivity distributions (Figure 6). Note that the high diffusivity values indicate the potential position of the chevron fractures.

For the inversion, the first 200 s of the recorded pressure data were utilized. No smoothing or filtering techniques were applied. The only restriction we applied was that we used only the first 200 s of each interference. The utilized transient pressure data are displayed separately in Figure 7 as well as Figures S1 and S2 in Supporting Information S1 for each inverted profile. Note in general all available interferences, that is, all interferences within the plane of the profile, are used for the inversion. However, all interferences connecting the upper and lower intervals along the borehole were judged with respect to potential short circuits along the packer (compare Section 3). Interferences characterized by very small response times (first arrival times) indicating the pressure pulses propagating along the borehole damaged zone, were excluded from the inversion.

5.4. Fracture Statistics

The fracture data were collected from the existing boreholes in the GER gallery. These include the nine vertical boreholes, which were used for the pneumatic tomography, that is, OHZ 6151–6159 as well as 20 other vertical boreholes along the whole GER gallery. In total 272 fractures were considered for the statistical analysis, which were grouped into two categories: (a) extension and (b) shear fractures. The spatial distribution of these fractures w.r.t. depth is shown in Figure 5a. According to Figure 5a, the extension fractures are mostly clustered in

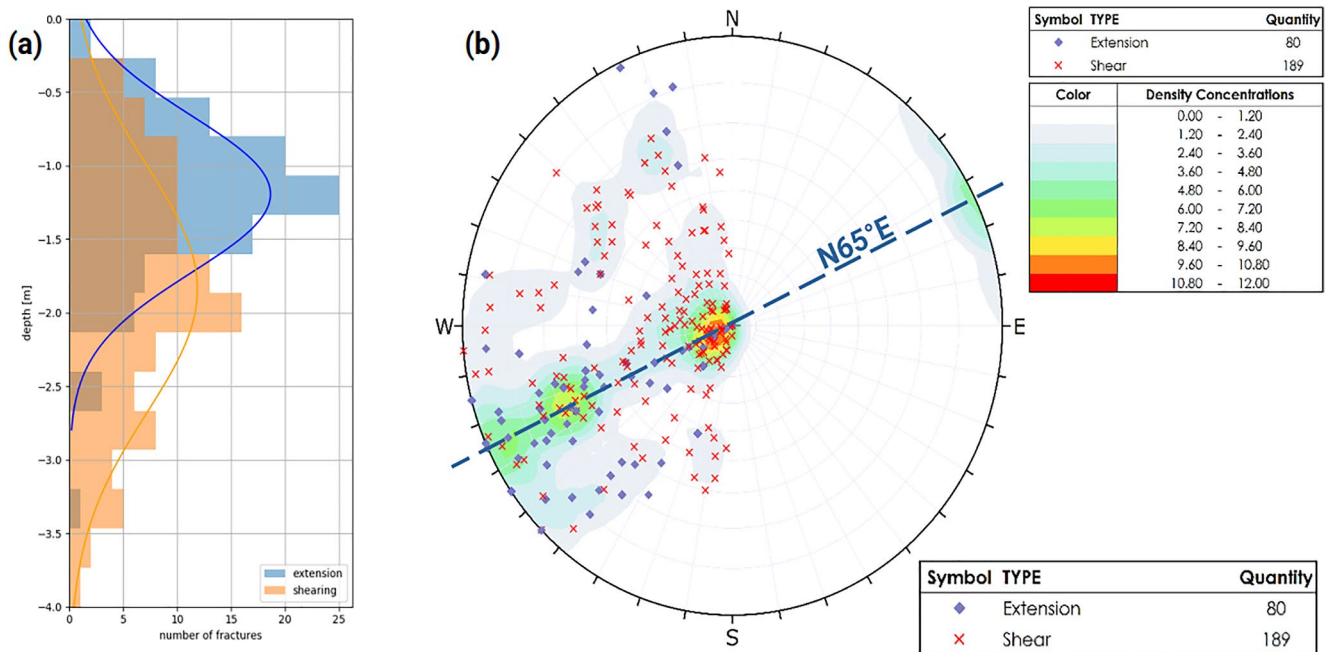


Figure 5. Fracture statistics: (a) fracture density of the considered fractures in vertical boreholes, (b) Stereonet plot of the poles for extension and shear fractures. The orientation of excavation is shown by the dashed blue line, that is, N65°E.

a depth above 2.8 m whereby the highest frequency of these fractures can be observed in a depth of 1.25 m. On the other hand, the shear fractures occur between 0 and 4.0 m with the highest frequency at a depth of 2.0 m. It is worthwhile to mention here that the collected fracture statistics data is based on core logging rather than geophysical image logs. In this case, the fracture statistics only from the GER 4 section might be subject to a statistical uncertainty of the fracture counts, especially at the depth close to the tunnel surface. This uncertainty is mostly due to poor recovery of the cores at highly fractured sections of EDZ. This is the main reason to consider 272 fractures from all the vertical boreholes along the GER gallery to compensate for the experimental bias as much as possible. In the following, the required fracture statistics, that is, fracture intensity, fracture orientation and fracture aperture, are defined.

5.4.1. Fracture Intensity

Fracture intensity refers to the number of fractures in a given rock mass and is often used as some terms such as *fracture spacing* ignore issues of orientation bias, spatial arrangement, and discontinuity size (Golder, 2016). Dershowitz and Herda (1992) define a class of fracture intensity measures in one, two, and three dimensions. In this work, we are using the following fracture intensities:

- P10: lineal fracture intensity, expressed as the number of fractures per unit length (L^{-1}).
- P21: areal fracture intensity, expressed as the length of fracture traces per unit area (L^{-1}).

For the DFN inversion, the areal fracture intensity (P21) is needed because the model domain of the utilized forward model is two-dimensional. Due to the limited diameter of the borehole cores, it is hard to derive such information from borehole mapping. In our study, we assume that $P21 = P10$. This is a reasonable assumption and the validity has been described in the literature (Barthélémy et al., 2009; Gauthier et al., 2012).

The calculated lineal fracture intensity (P10) for all the vertical boreholes in the GER section is about 2.0. In this case, an areal fracture intensity of $P21 = 2.0$ is used for the DFN inversion. Fracture intensity is used by the inversion first at the generation of the initial DFN. At later iterations, fracture intensity acts as a limiting factor, if it deviates too far from the defined value the probability of fracture addition or deletion is set to zero.

5.4.2. Fracture Orientation

The geometrical information of 272 fractures including the dip and dip direction values was collected and plotted in a rose diagram (Figure 5). The dominant dip direction angle is N65°E, which is the excavation direction

Table 2
Equivalent Two Dimensional Hydraulic Aperture Used in DFN Inversion

Model	Fracture	Profile 1 (mm)	Profile 2 (mm)	Profile 3 (mm)
Upper model domain	Extension fracture	3.0	3.0	1.89
	Shear fracture	3.2	1.28	0.89
Lower model domain	Extension fracture	0.3	0.3	0.3
	Shear fracture	0.32	0.32	0.33

of gallery GER (Figure 5). This shows that the conceptual model proposed by Armand et al. (2014) describes nicely the geometrical characteristics of the excavation-induced fractures around the gallery (Figure 2). The extension and shear fractures in Figure 5 are shown with blue diamond and red cross symbols, respectively. As we expected from the conceptual model and shown on the 3D diffusivity tomogram, the majority of the extension fractures are sub-vertical, whereas the shear fractures are sub-horizontal. These data helped us to define the range of dip angles for the 2D inversion of the induced fractures orthogonal (i.e., the profile is parallel) to the tunnel axis (excavation direction). Based on these data, the dip angles of 80° and 40° for the extension and shear fractures are considered, respectively. Based on these angles, two fracture sets are defined for the inversion process.

5.4.3. Fracture Aperture

The gas conductivity values estimated from the pneumatic injection tests conducted in these OHZ boreholes and analyzed using the borehole simulator MULTISIM as well as the reconstructed diffusivity values were used to appraise the range of fracture apertures (Table 1). The derived apertures of extension and shear fractures using Cubic's law are in the range of 0.02–0.04 mm and 0.003–0.009 mm for the extension and shear fractures, respectively. It is important to mention that the calculated hydraulic aperture using Cubic's law is derived from the analysis of pressure responses recorded in a 3D environment, however, the applied DFN inversion is based on a 2D model domain. Hence, it is necessary to scale the derived apertures or transform the aperture into an *equivalent 2D hydraulic aperture* to honor the 2D model domain. Various authors such as Preisig et al. (2012) have tried to extract these scaling factors. Based on their approach the flow rate needs to be reduced by roughly two orders of magnitude to be able to use the flow rate as the boundary condition for 2D modeling. In this study, we use the same order of magnitude enhancement on fracture hydraulic aperture and then apply the injection pressure boundary conditions to simulate the hydraulic tomography tests. To do so, we use an *equivalent 2D hydraulic aperture* in the range of 0.3–4.4 mm. Finally, the aperture was manually fitted—separately for each profile—by the manual fitting of selected single interferences. The finally applied *equivalent 2D hydraulic apertures* for the different DFN test series are given in Table 2.

5.5. Forward Model

The forward model is based on the conservation of fluid mass for a control volume without any chemical reaction, which allows the simulation of the pressure propagation inside the fracture network (Jalali, 2013). This model is based on the implicit finite difference method approach, which allows for resolving both steady-state and transient pressure propagation inside the fractures. In this work, only the fracture network in the considered domain is discretized and it is assumed that the matrix is impermeable. This assumption is an approximation, but appropriate for short-term, low-pressure gas injection into the Callovian Oxfordian formation. By simulating a vertical cross-section between the OHZ boreholes, the thickness of fractures is assumed to be constant (perpendicular to the 2D DFN plane), and fracture aperture and length control the storativity of each fracture element. Fracture conductivity for each element is calculated using the cubic's law, assuming each fracture segment consists of two parallel plates (fracture walls).

In total three 2D DFN planes were selected and used for the DFN inversion which are parallel to the GER tunnel axis (Profiles 1–3). For each profile, all observed pressure interferences were simulated and during the inverse step compared with the measured pressure interferences by calculating the root mean square residual (RMS) for all interferences associated with the respective profile. The following boundary conditions (BCs) are applied to the forward model:

- Constant atmospheric pressure is applied to all the fracture elements with a depth less than 0.5 m from the tunnel surface
- The depth dependence of the *equivalent 2D hydraulic aperture* is introduced in the model as follows: a fracture boundary is defined at the depth of 2.8 m to account for the decrease of the *equivalent 2D hydraulic aperture* with increasing fracture depth in the forward model. All fractures above the fracture boundary are characterized by a larger *equivalent 2D hydraulic aperture* than the fractures below the boundary. If a fracture

boundary intersects the boundary in a depth lower than 2.8 m, the upper part of the fracture is assigned with a larger *equivalent 2D hydraulic aperture* than the lower part. The relationship between the *equivalent 2D hydraulic aperture* above and below the boundary can be described by a constant *equivalent 2D hydraulic aperture* multiplier in the range between 1/5 and 1/10 (compare Table 2).

- The gas injection pressure measured at each source interval during the HT experiment is used as the injection boundary condition for the forward model. This includes the first 200 s of gas injection for each test. A sampling frequency of 1 s (i.e., 200 data points) is used without any data smoothing.
- The fluid flow is only modeled in the fracture network and it is assumed that the rock matrix is impermeable, so the BCs between fractures and matrix is no-flow BC.
- The observed pressures at the monitoring intervals are simulated for 200 s utilizing the described forward model and finally compared with the measured pressure response at the respective monitoring intervals.

The other used physical properties for the forward model are as below:

- Gas dynamic viscosity: 1.5×10^{-5} Pa.s
- Gas compressibility: 1.0×10^{-6} 1/Pa
- Gas density (at 20°C): 1.1606 kg/m³

5.6. Results—Fracture Probability Maps

Figure 6 shows slices—derived from the 3-D diffusivity tomogram—parallel to the excavation direction/tunnel axis. The arrow in Figures 6a and 6b indicates the excavation advance direction. The slices display the exact position of the 2D inversion results and are utilized to interpret the reconstructed fracture probability maps. The gray dashed line within the diffusivity tomograms indicates the assumed trace of the chevron fracture. Note, within the conceptual, model (Figure 2) the chevron fractures are displayed in brown color. In the following, we will present the reconstructed 2D fracture probability maps and compare them to the results of the travel time-based inversion approach.

A standard way of presenting the result of Bayesian inversion is via calculating the model average over the chain. In the case of the DFN inversion, this is done by producing a fracture probability map (Somogyvári et al., 2017). The fracture probability map is calculated via the rasterization of the DFN realizations and by taking the mean of these raster models over the ensemble. The fracture probability at one pixel, that is, the probability of this position contains a fracture, is the number of DFN realizations that identify a fracture at this point divided by the size of the ensemble.

Figure 6 shows the three 2D fracture probability maps. By comparing the fracture probability maps with each other, it became obvious that all the profiles show a similar fracture pattern. Particularly, the vertical inclined fracture connecting the lower and upper horizontal shear fractures are at the same spatial position (lower end at x-position approx. 80 cm) and the fracture location is associated with a high probability as can be seen from the red color. The vertical incline extension fracture is an integral part of the chevron structure. The fact that this structure is located in each profile at the same x-position would allow the interpolation of the fracture in 3D. The 3D interpolation would mimic the 3D shape of the chevron structure shown in the conceptual model in brown displayed in Figure 2. These similarities are an indication of the robustness of the methods because the reconstructions were generated in independent model runs.

It can be observed that the position of the upper horizontal shear fracture is associated with a high probability in all three profiles. This is also in accordance with the conceptual model and the diffusivity tomograms. Both show for the upper part a larger permeability indicating a high probability for a direct hydraulic connection of the boreholes by shear fractures. Additionally, it can be observed that for Profiles 1 and 2 fracture connections to the tunnel surface—associated with a high probability—were reconstructed. Note the constant head boundary simulating atmospheric pressure was implemented 0.5 m below the tunnel surface (y-axis = 450 cm). The existence of the extension fractures intersecting the tunnel surface could be observed during pneumatic test execution. The reconstructed shear fractures (or their spatial location, respectively), in the lower part of Profiles 1 and 2 are associated with a lower probability and show a larger range of possible fracture connections. This indicates that the fracture connection in the lower part is less significant in comparison with the upper part of the profiles. Profile 3 located in the central part of the tunnel, however, also shows a clear fracture connection (associated with a high

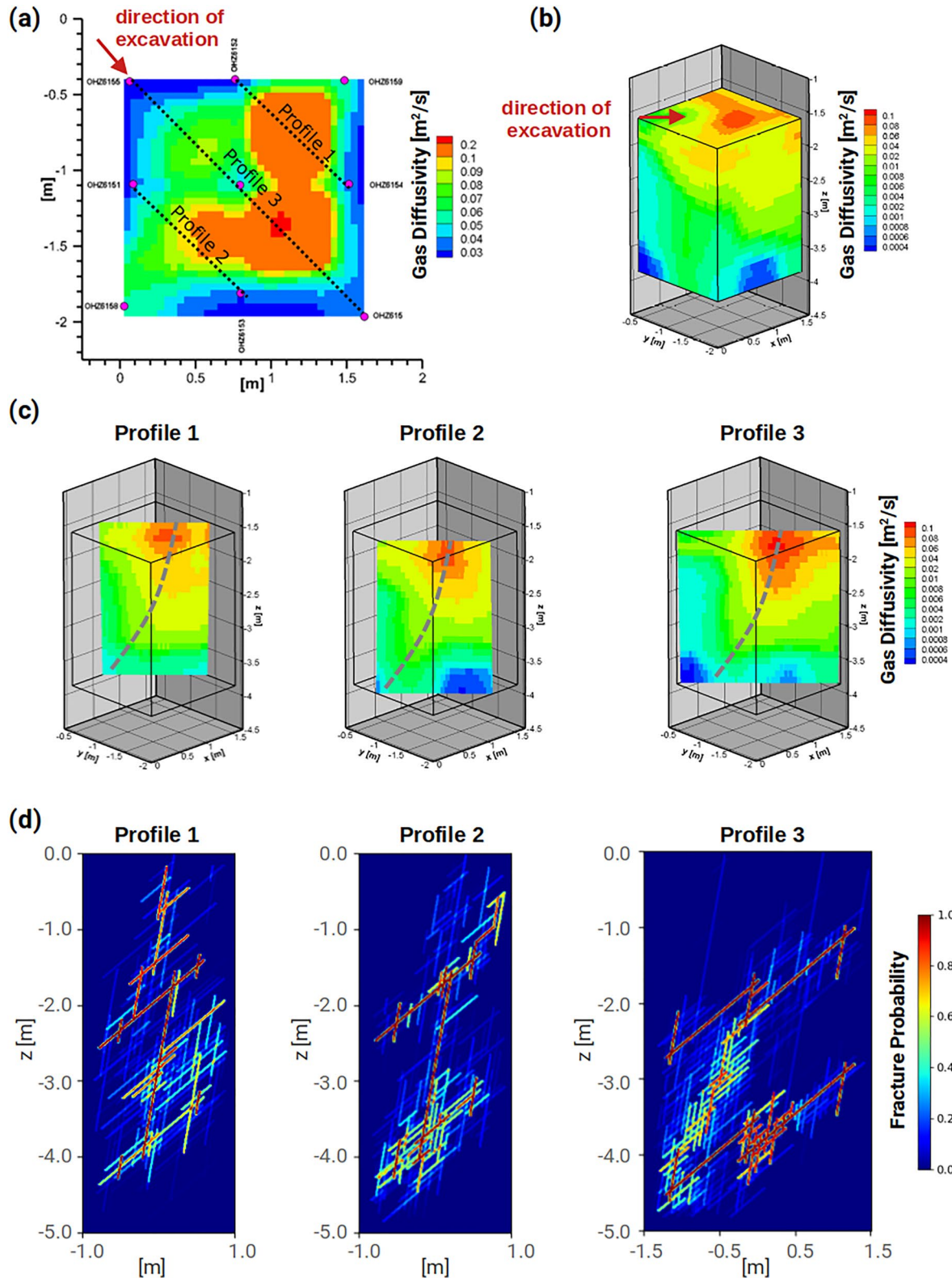


Figure 6. Comparison between travel time-based and discrete fracture network-based inversions: (a) horizontal slice of the reconstructed diffusivity model showing the three profiles used for DFN inversion, (b) 3D model domain with the arrow indicating the tunnel axis direction, (c) vertical slices parallel to the tunnel axis which correspond to profiles 1–3. The dashed gray line represents the conceptual chevron fractures, (d) fracture probability maps of the Profiles 1–3.

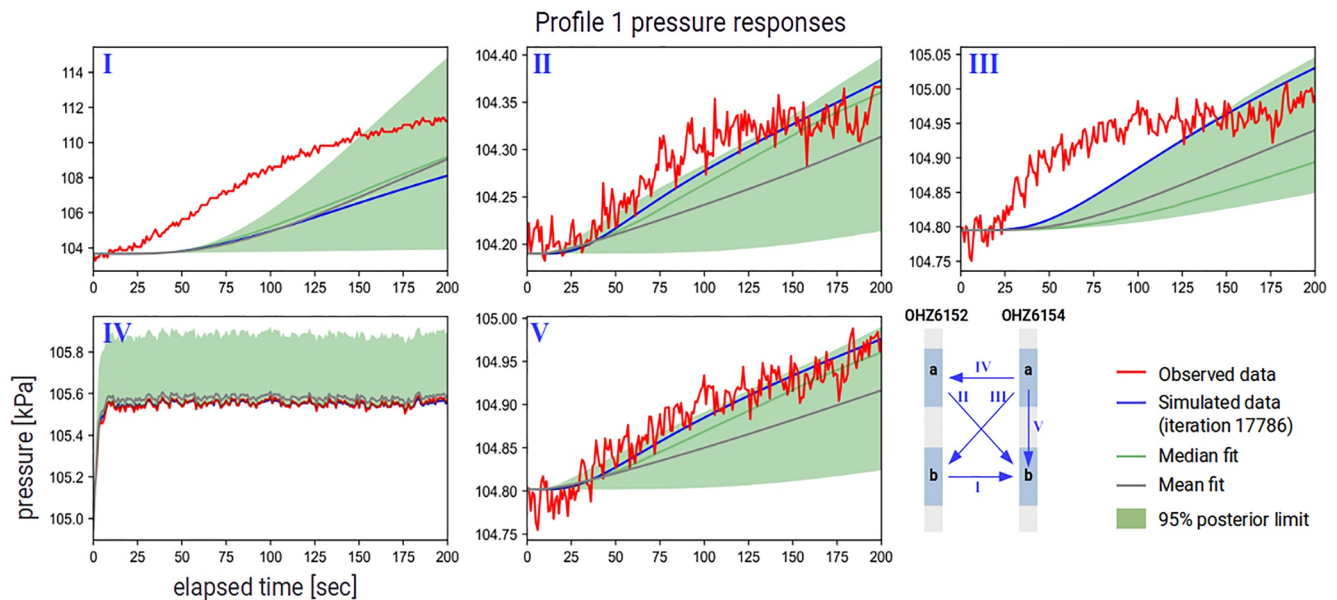


Figure 7. Comparison of observed pressure responses with the DFN inversion in Profile 1. The direction of injections was shown by Roman numbers, (e.g., I) corresponding to the pressure response in OHZ6154-b interval due to injection into OHZ6152-b interval. In this figure the red line is the observed (measured) data, the blue line is the simulated data @ iteration 17786 and the gray and green lines are the mean and median fits of all the iterations, respectively. The green area shows 95% of the posterior data. The horizontal axis gives the elapsed time in seconds whereas the vertical axis gives the pressure value in kPa.

probability) between the three boreholes. This might be explained by the fact that the induced fracture network in the center of the tunnel is stronger developed (penetrates deeper) than at the edges of the tunnel.

In Figure 7, the comparison between measured and simulated pressure response curves for Profile 1 is shown. For the simulated pressure responses, the mean and median fits of all the iterations (gray and green lines, respectively), 95% of the posteriors (green area) and the last accepted iteration (iteration 17786 for Profile 1 in blue line) are shown. The comparison between measured and simulated pressure response curves show that the DFN model captures the characteristic of the induced fracture network. Only the amplitude of the interference recorded in the lower part of the model domain (marked I) is underestimated by two kPa and the early times of the interference connecting the upper and lower zone (marked III) could not be captured perfectly. The three interferences (marked by II, IV, and V) can be matched with high quality. Particularly, the horizontally aligned interference in the upper part of the model domain (marked IV) is fitted with extremely high quality. Even the small pressure variation in the range of a few pascals originating from a slide variation of the injection pressure/rate could be reproduced by the mode.

6. Summary, Discussion, and Conclusion

The goal of the study is to reconstruct the hydraulic properties and the shape of the induced fracture network in order to improve the insights and validate the conceptual model of the fracture network as a function of the stress field and to show the potential of tomographic measurement arrays in combination with smart analysis/inversion techniques. Within the presented study, the cross-hole responses of the pneumatic injection tests, recorded during the OHZ/GER experiment performed in the Meuse/Haute-Marne Underground Research Laboratory (URL), located in the Callovo-Oxfordian clay rock formation, were analyzed with a combination of an equivalent porous media approach and a discrete fracture network inversion approach. For the EPM approach, we utilized a hydraulic tomographic travel time approach and for the DFN approach, we applied a transdimensional reversible jump Markov Chain Monte Carlo algorithm, whereby the results of the travel time-based inversion approach are used to constrain and interpret the DFN inversion results.

The tomographic inversion is based on 18 gas injection tests and several hundred pressure interferences, which were recorded between nine closely spaced boreholes in the tunnel floor. Each borehole is equipped with a multi-packer system consisting of two intervals that can be used for injection and/or observation, respectively.

In all 18 intervals, constant rate injection tests were performed. For the travel time-based inversion approach, 151 interferences were utilized to reconstruct a 3D diffusivity tomogram. The 3D diffusivity tomogram allows reconstruction of the trace of t a deeply inclined structure, which dips in the opposite direction of the excavation direction. These structures can be interpreted in accordance with the conceptual model developed by Armand et al. (2014) as lower chevron structures. The model of Armand et al. (2014) predicts that chevron structures are initiated ahead of the excavation face during excavation and form symmetrically to the horizontal plane crossing the gallery axis. Finally, it has to be highlighted that the reconstruction of moderately inclined structures is hardly possible with conventional hydraulic test analysis techniques because the spatial assignment of the reconstructed parameter estimates is non-unique. However, the reconstructed 3D diffusivity tomogram did not show clearly the trace of the lower chevron structure, but a blurry picture with increased diffusivity values, where we assume the position of this structure. In order to improve the significance of the tomographic inversion results, we have used the outcome of the travel time-based tomographic approach to constrain the DFN inversion performed in the second step.

For the DFN inversion, we reconstructed three 2D profiles which were aligned parallel to the excavation direction. It could be shown by comparing the three reconstructed 2D fracture probability maps with each other that all profiles show a similar fracture pattern. Particularly, the vertical inclined fracture connecting the lower and upper horizontal shear fracture is at the same spatial position and is associated with a high probability. The fact that this structure is located in each profile at the same spatial position would allow interpolating the fracture in 3D. The 3D interpolation would display the 3D shape of the chevron structure shown in the conceptual model of Armand et al. (2014) and improves strongly the significance of the 3D diffusivity tomogram.

The comparison between measured and simulated pressure response curves show that the DFN model captures the characteristic of the induced fracture network. Most interferences can be matched with relatively high quality. Only the interferences located in the lower part of the tomogram show deviations between simulated and measured pressure interference. There are several potential reasons for the observed deviations:

1. The amplitude of the pressure response in the lower part is very small, partly smaller than 1 kPa, and hence, the data quality can be influenced by the limitations of the experimental setup, for example, the limitations of pressure transducers. Note the performance of gas interference tests in low permeability media such as the Callovian Oxfordian Clay rock requires excellent equipment and experienced personnel.
2. The forward model is a strong simplification of reality, particularly, with respect to fracture aperture. The depth dependence of the *equivalent 2D hydraulic aperture* is introduced in the model by means of a fracture boundary in a depth of 2.8 m. All fractures above the fracture boundary are characterized by a larger *equivalent 2D hydraulic aperture* than the fractures below the boundary. If a fracture boundary intersects the boundary in a depth lower than 2.8 m, the upper part of the fracture is assigned with a larger *equivalent 2D hydraulic aperture* than the upper part of the fracture is assigned with a larger *equivalent 2D hydraulic aperture* than for the lower part. The fracture aperture is a very sensitive parameter and was not fitted within the parameter optimization procedure. However, in this context, it has to be mentioned that the hydraulic interference signal of one large fracture and several small fractures is identical as long as the volumes of the large fracture and the sum of the volume of the small fractures are identical. Hence, the algorithm will overcome the challenge of an *equivalent 2D hydraulic aperture* which was estimated mistakenly slightly too small by generating several fractures that are aligned parallel to each other. If the fracture aperture is chosen mistakenly too large the algorithm will increase the *length of the fracture network* between the injection and observation interval by moving, generating and deleting fractures. Generally, hydraulic signals alone are not able to overcome the non-uniqueness associated with the fracture aperture. A possibility to overcome the non-uniqueness would be the integration of transport information into the inversion which we judge as a worthwhile area of further research.
3. The forward model comprises only two fracture sets with a dip angle of 80° and 40°. The utilization of only two fracture sets is a simplification of the induced fracture network observed around the gallery. However, a larger number of fracture sets would increase the non-uniqueness during inversion and would lead to an ill-posed inverse problem associated with high uncertainty. The only way to overcome this dilemma would be the incorporation of additional information.
4. The applied forward model treats the matrix as impermeable, which is a simplification because the clay matrix is characterized by a low permeability. The matrix contribution is increasing with decreasing fracture

density and/or decreasing fracture aperture. This effect might be an explanation for the relatively poor fit of the interferences recorded in the lower part of the model domain, characterized by smaller fracture apertures than the upper part of the model domain.

Although we listed several limitations associated with the proposed inversion scheme, we would like to highlight that the presented inversion results allowed us to validate the conceptual model of the excavation-induced fracture network for the ceiling and floor of the gallery (Armand et al., 2017). The reconstructed tomograms and fracture probability maps provide information about fracture connectivity, hydraulic properties and the visual reconstruction of the dominant structures, which goes far beyond the expressiveness of conventional hydraulic test analysis. The combined interpretation of the diffusivity tomograms and the fracture probability maps provide a high system understanding.

The synthesis of the results between the reconstructed 3D diffusivity tomogram and the 2D fracture tomograms improved the insights in the spatial geometry of the induced fracture network around galleries: the gas tomographic travel time-based approach allowed for the spatial appraisal of the moderately inclined chevron structures in 3D and the 2D fracture tomograms provide spatially high-resolution images of the fracture traces associated with the chevron structures.

Data Availability Statement

The interferences data used for the EPM and DFN inversions in the study are available online via <https://doi.org/10.18154/RWTH-2022-09250>.

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